

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech CSE**

Semester: 6th

Name of the Course: Image Processing and Computer Vision

Course Code: TCS 691

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1

- (a) Compare and contrast low-level, mid-level, and high-level vision. Provide examples of tasks or applications that fall under each category. (10 Marks) CO1

OR

- (b) Write a MATLAB/Octave script that reads an image file, converts it to grayscale, and displays the original and processed images side by side. (10 Marks) CO2

Q2

- (a) Explain the significance of histogram equalization in enhancing image contrast. Demonstrate the process with an example. (10 Marks) CO2

OR

- (b) Discuss the concept of gamma correction. Provide a real-world scenario where gamma correction is useful in image processing. (10 Marks) CO2

Q3

- (a) Explain the concept of morphological operations in image processing. Discuss the applications of erosion and dilation operations with the help of examples. (10 Marks) CO2

OR

- (b) How does spatial filtering work in image processing? Apply a 3x3 averaging filter on a 5x5 image matrix and show each step of the process. (10 Marks) CO1

Q4

- (a) Perform edge detection using the Prewitt operator on a given 5x5 image matrix. Show all steps in detail. (10 Marks) CO2

OR

- (b) Discuss the role of median filtering in noise reduction. Apply a 3x3 median filter to a 5x5 noisy image matrix and demonstrate each step. (10 Marks) CO2

Q5

- (a) Show an analysis of Image compression using Huffman code by considering a suitable example. (10 Marks) CO2

OR

- (b) Find within class variance using Otsu's thresholding for below matrix by considering a single threshold value. CO3

(b)
$$\begin{bmatrix} 6 & 5 & 4 & 3 & 2 & 12 \\ 5 & 4 & 3 & 2 & 12 & 2 \\ 4 & 3 & 2 & 12 & 2 & 3 \\ 3 & 2 & 12 & 2 & 3 & 4 \\ 2 & 12 & 2 & 3 & 4 & 5 \\ 12 & 2 & 3 & 4 & 5 & 6 \end{bmatrix}$$

